Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 15)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Isle of Eigg (area 31 km^2, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 1 . 1 Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

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0 1 . 2 Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

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0 1 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

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0 1 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

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0 2 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 2 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

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0 2 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

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0 2 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

0 2 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

0 2 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

0 3 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 3 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 3 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

0 3 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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0 4 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 4 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 4 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 5 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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2 .....

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0 5 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 5 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

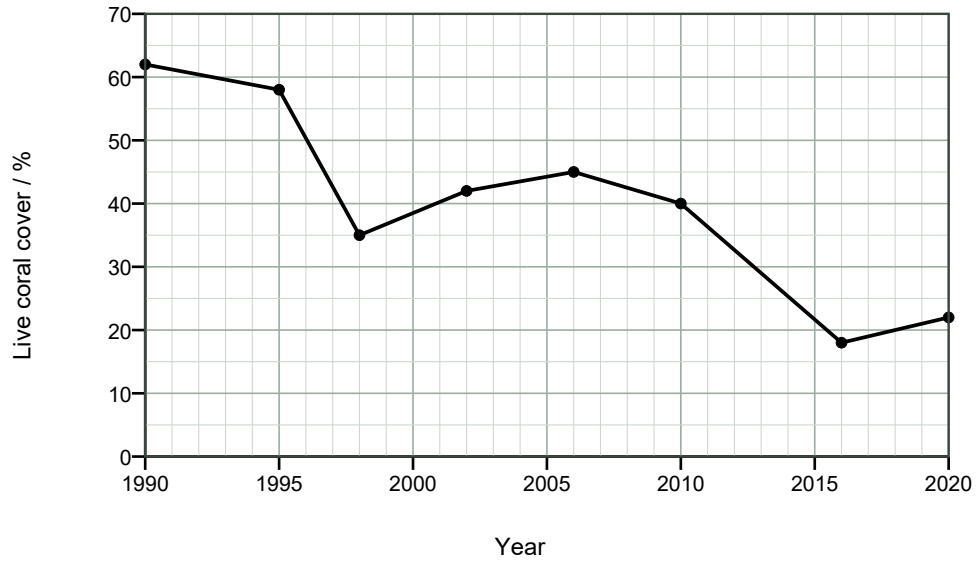
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0 6 Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 6 . 1 Describe the changes in coral cover shown in Figure 1.

[2 marks]

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0 6 . 2 Explain what coral bleaching is and give **three** causes.

[4 marks]

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0 6 . 3 Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

0 6 . 4 Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

10

0 7 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 7 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 8 Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 8 . 1 Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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0 8 . 2 Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

0 8 . 3 Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

0 8 . 4 Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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0 9 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 9 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 9 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 9 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

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2 .....

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0 9 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ..... %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

1 1 . 2 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS